9.4 Experimental Design

Name _____ Period ____ Date _____

Algebra II 9.4 (30-minute plan) · Student sheet

I can…

- describe a randomized comparative experiment and identify the treatment and control groups.
- tell an experiment from an observational study.
- identify problems in an experiment's design.

Vocabulary · fill in the missing word

Controlled experiment — compares a _____ group with a control group

Treatment group — the group that _____ the treatment

Control group — the group that does not (it may get a _____)

Placebo — a harmless, _____ treatment that looks like the real one

Randomized comparative experiment — subjects are randomly assigned to the _____ or control group

Replication — repeating an _____ with many subjects so results aren't due to chance

Observational study — observes subjects without _____ a treatment; it can't show cause and effect

Part 1 · Notes and examples (fill in each note, then work the example)

A good experiment: A good experiment has a _____ group and a _____ group, _____ assignment, and enough subjects (_____).

Example 1

A company tests a new energy drink. 100 volunteers choose whether to drink it or water before a run, and their times are compared. Is this a randomized comparative experiment? Explain.

Example 2

60 patients are randomly assigned to take a new sleep pill or a placebo, and their hours of sleep are compared. Identify the treatment group and the control group.

Cause and effect: Only an _____ can show cause and effect.

Example 3

Tell whether each is an experiment or an observational study.

a. Researchers compare the test scores of students who eat breakfast with those who don’t.

b. A teacher randomly assigns half of a class to study with flashcards.

Part 2 · Partner practice

Partner practice

A study records how many hours teens sleep and their grades. Is it an experiment or an observational study? Can it show that sleep causes better grades?